cubic meter might not sound like a lot, there are many cubic meters of space out there in the cosmos. Moreover, the farther back in time you look, the smaller the space into which the mass/energy is squeezed, so the denser the universe becomes.

23 Even though a decrease in symmetry means that fewer manipulations go unnoticed, the heat released to the environment during these transformations ensures that overall entropy—including that of the environment—still increases.

24 The terminology isn't particularly important, but briefly, here's where it comes from. The valley in Figure 9.1c and 9.1d has a symmetric shape—it's circular—with every point being on a par with every other (each point denotes a Higgs field value of lowest possible energy). Yet, when the Higgs field's value slides down the bowl, it lands on *one* particular point on the circular valley, and in so doing "spontaneously" selects one location on the valley as special. In turn, the points on the valley are no longer all on an equal footing, since one has been picked out, and so the Higgs field disrupts or "breaks" the previous symmetry between them. Thus, putting the words together, the process in which the Higgs slides down to one particular nonzero value in the valley is called *spontaneous symmetrybreaking*. Later in the text, we will describe more tangible aspects of the reduction of symmetry associated with such a formation of a Higgs ocean.[7]

25 You might think I've left out an "i" in the last syllable of "inflaton," but I haven't; physicists often give fields names, such as photon and gluon, which end with "on."

26 As the universe expands, the energy loss of photons can be directly observed because their wavelengths stretch—they undergo *redshift*—and the longer a photon's wavelength, the less energy it has. The microwave background photons have undergone such redshift for nearly 14 billion years, explaining their long—microwave—wavelengths, and their low temperature. Matter undergoes a similar loss of its kinetic energy (energy from particle motion), but the total energy bound up in the mass of particles (their *rest energy*—the energy equivalent of their mass, when at rest) remains constant.